Software Verification & Validation (V&V)

RCS 305 ? Software Engineering | Riara University Revision Resource

1. The Foundational Duality (Boehm)

? Verification: ’Are we building the product right?’ Confirms that software artifacts comply with specifications established in preceding development phases.
? Validation: ’Are we building the right product?’ Confirms that the realized software satisfies actual user needs and business requirements.

2. Static vs. Dynamic Techniques

? Static Verification (Non-executable): Formal inspections, peer code reviews, walkthroughs, linters, and formal mathematical proofs of correctness.
? Dynamic Validation (Executable): Running test suites against compiled binaries with concrete input vectors to observe actual runtime responses.

3. Formal Technical Reviews (FTR)

Structured meetings conducted by independent peers to detect functional errors, logic discrepancies, and standards violations before artifact sign-off.

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